

CO5-AT3-Level Exam3

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QUESTIONS

LEVEL 3 — HARD

Q1
Numbers**Q2**
Control

Level 3 — Hard (H) Number Palindrome — Numbers

✓ Submitted

Develop a code to Reverse and Add a Number until you get a Palindrome.

Sample Input If 7325 is input number, then 7325 (Input Number) + 5237 (Reverse Of Input Number) = 12562

12562 + 26521 = 39083

39083 + 38093 = 77176

77176 + 67177 = 144353

144353 + 353441 = 497794 (Palindrome)

Testcase: 1. 8765

2. -8765

3. 0

4. EIGHT FIVE

5. 87.57

2 / 2 saved

Your Code

```
1 import java.util.*;
2 public class Main{
3     static int reverse(int n){
4         int r=0;
5         while(n>0){
6             r=r*10+n%10;
7             n=n/10;
8         }
9         return r;
10    }
11
12    static boolean palindrome(int n){
13        return n==reverse(n);
14    }
15
16    public static void main(String[] args){
17        Scanner sc=new Scanner(System.in);
18        int n=sc.nextInt();
19        while(!palindrome(n)){
20            n=n+reverse(n);
21        }
22        System.out.println("Palindrome : "+n);
23    }
```

Output

Palindrome : 497794

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QUESTIONS

LEVEL 3 — HARD

Q1
Numbers

Q2
Control

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Level 3 — Hard (H) Steps — Control

Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

Example 1:

Input: num = 14

Output: 6

Explanation:

- Step 1) 14 is even; divide by 2 and obtain 7.
- Step 2) 7 is odd; subtract 1 and obtain 6.
- Step 3) 6 is even; divide by 2 and obtain 3.
- Step 4) 3 is odd; subtract 1 and obtain 2.
- Step 5) 2 is even; divide by 2 and obtain 1.
- Step 6) 1 is odd; subtract 1 and obtain 0.

Testcase:1. n = 5

2. n = 10

3. n = 12

4. n = 18

5. n = 20

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Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int number=sc.nextInt();
6         int steps=0;
7         while(number>0){
8             if(number%2==0)
9                 number=number/2;
10            else
11                number=number-1;
12            steps++;
13        }
14        System.out.println("Number of Steps: "+steps);
15    }
16 }
```

Output

```
Number of Steps: 6
```